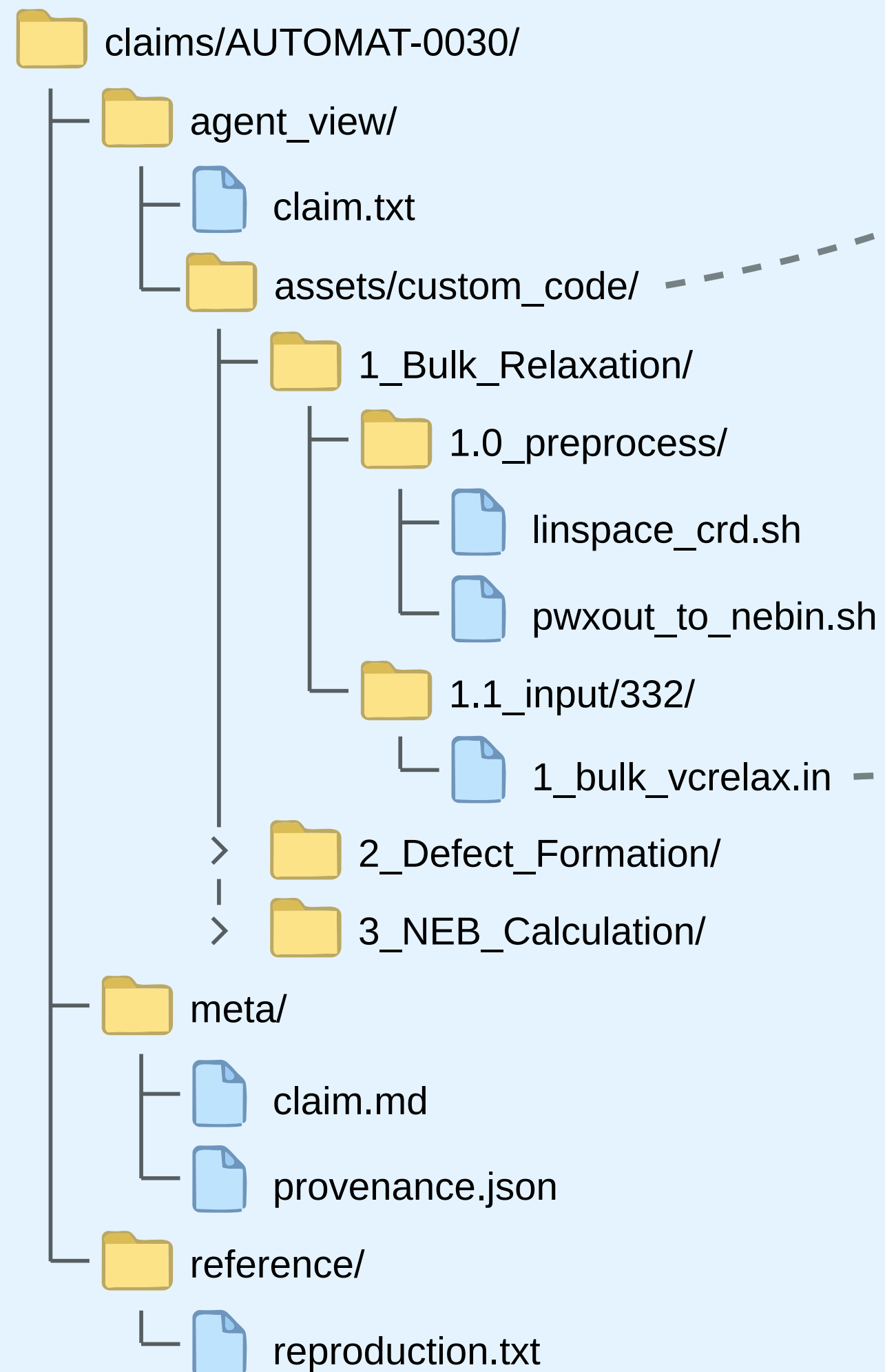


Reproduction Package



Claim Detail

AUTOMAT-0030

Low-energy pathways lead to self-healing defects in CsPbBr₃

DOI: <https://doi.org/10.1039/d5cp01641j>

The defect formation energy (E_f) for a neutral Br_{Pb} is calculated to be 1.30 eV, close to the value of 1.27 eV for the negatively charged state at VBM.

Subdomain: Density Functional Theory (DFT)

Type: From-paper reproduction

Tools: Python/QuantumEspresso

reproduction.txt

1. Required input files: CsPbBr₃ supercell, pseudopotentials...
2. Create BrPb anti-site defect in both neutral and negatively charged (-1) states
3. ...

```
&CONTROL
  calculation = 'vc-relax'
  prefix      = 'CsPbBr3_bulk'
  pseudo_dir  = './pseudo/'
/
ATOMIC_POSITIONS {angstrom}
Cs   0.000000  0.000000  0.000000
Pb   0.500000  0.500000  0.500000
...
```